

DEGREE: MSc in Artificial Intelligence

Module: Introduction to Artificial Intelligence

Assignment Title: Predictive Modelling for Credit Card Default Using Machine Learning

Assignment Type: Practical skills assessment

Word Limit: 1000-1500 words

Weighting: 50%

Issue Date: 24/04/2026

Submission Date: 10/06/2026

Feedback Date: 01/07/2026

Issued by: Dr. Abdelaziz Triki

Plagiarism:

When submitting work for assessment, students should be aware of the InterActive/Canvas guidance and regulations in concerning plagiarism. All submissions should be your own, original work.

You must submit an electronic copy of your work. Your submission will be electronically checked.

Learner declaration		
Word count		
Use of proof-reader/proof-reading service (see guidance contained in Academic Misconduct Regulations on appropriate use)	YES	NO

If 'yes', please specify which service (e.g. Grammarly, Studiosity etc.)		
I confirm that I submit this work as my own work and that I have cited all sources I have used, and I understand that using sources without citing them correctly may be considered as Academic Misconduct.	YES	NO
I confirm that I have followed guidance on the acceptable use of AI tools for this assignment where such guidance has been issued by my tutors.	YES	NO
Where use of appropriately cited use of AI tools is permitted, I confirm that I have cited in accordance with the UCA Harvard Referencing Standard.	YES	NO

Harvard Referencing:

The Harvard Referencing System must be used. The Wikipedia, UKEssays.com or similar websites must **not** be used or referenced in your work.

By submitting this assessment, you are confirming that you have understood and are complying with the University's Academic Misconduct Regulations.

Plagiarism Detection (Turnitin)

BSBI uses Turnitin (an on-line text matching tool) to help identify cases of plagiarism. Turnitin will compare a submitted assignment with a database of journal articles, books, web pages and previously submitted assignments.

Signature (Student): _____ Date: _____

Introduction

Credit card default poses a significant challenge for financial institutions around the world. When cardholders fail to meet their payment obligations, banks and issuers incur losses and increased risk. Machine learning enables data-driven risk assessment by leveraging demographic, behavioral and financial variables to predict the likelihood of default. Understanding which clients are likely to default can inform credit decisions and allow for proactive risk mitigation.

This assignment requires you to design, evaluate and interpret multiple machine-learning models using a large dataset to predict whether a credit card client will default on their payment.

Learning Outcomes:

LO1. Demonstrate the ability to design and execute AI experiments, showing innovative problem-solving skills and adapt AI techniques to new challenges

LO2. Conduct in-depth research on AI trends and technologies, analyze the latest advancements, and assess potential impact on society

LO3. Engage in a collaborative real problem-based projects using modern AI tools

Assessment Criteria: Weighting 50%

1000-1500 words

Dataset

Use the provided dataset: Credit_Card.csv (34,788 records, 30 features). The dataset is provided directly as part of this assignment. No external download is required.

Link: <https://github.com/abdelDebug/Predictive-Modelling-for-Credit-Card-Default-Using-Machine-Learning.git>

The dataset contains demographic, credit, payment history, and risk-related variables for credit card clients, covering a six-month period (April–September 2005). The target variable **default.payment.next.month** indicates whether a client defaulted on their payment in the following month (1 = Yes, 0 = No). Additional columns include a city label (CITY), a risk classification (RISK_RATING), a cumulative billing summary (BILL_AMT_SUM), a log-transformed credit limit (LIMIT_BAL_LOG), and a continuous risk score (risk_leak). Note that the dataset contains **missing values** in several columns, which must be addressed during data preparation.

Tasks (All Tasks Are Equally Weighted)

Task 1: Exploratory Data Analysis (20%)

- Investigate the distribution of the target variable `default.payment.next.month` and discuss the class imbalance observed.
- Generate descriptive statistics for key numerical features (mean, median, standard deviation), including `LIMIT_BAL`, `AGE`, `BILL_AMT_SUM`, `LIMIT_BAL_LOG`, and `risk_leak`.
- Create visualisations (histograms, bar charts, box plots) to identify trends, outliers, and skewness across numerical and categorical features.
- Explore the distribution of categorical features such as `SEX`, `EDUCATION`, `MARRIAGE`, `CITY`, and `RISK_RATING`.
- Explore correlations between numerical features and the target variable.
- Identify features that appear most strongly linked to default risk, paying particular attention to `risk_leak`, `RISK_RATING`, and payment history columns (`PAY_0` to `PAY_6`).

Task 2: Data Preparation (20%)

- Identify all missing values in the dataset. Then, describe and justify your chosen handling method (e.g., imputation or row removal) for each.
- Encode categorical variables appropriately if needed.
- Apply scaling to numerical variables (StandardScaler, MinMaxScaler, or similar).
- Justify each preparation step and provide before-and-after examples where possible.

Task 3: Model Training (20%)

- Split the dataset into training and testing sets. Justify your chosen split ratio and discuss the class imbalance in the target variable.
- Train multiple classification algorithms (at least five), such as:
 - Logistic Regression
 - Support Vector Machine
 - Decision Tree
 - Random Forest
 - K-Nearest Neighbors
 - Gradient Boosting / XGBoost (*optional*)
- Justify your algorithm selection in relation to the dataset characteristics (size, feature types, class imbalance) and the prediction objective.
- Train each model on the training set using default hyperparameters.

- Clearly record all initial hyperparameters before any tuning is performed.

Task 4: Model Evaluation and Visualisation (20%)

- **Evaluate all trained models using appropriate metrics:** accuracy, precision, recall, F1-score, and AUC-ROC.
- Present results in a clear comparative table.
- **Provide visual diagnostics:** confusion matrices and ROC curves (one-vs-rest or macro-averaged) for each model.
- Select the **best-performing model(s)** and perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV.
- Compare model performance before and after tuning.
- Use SHAP, LIME, or similar methods to interpret the best model's predictions. Discuss the relative importance of features such as risk_leak, RISK_RATING, PAY_0, LIMIT_BAL, and BILL_AMT_SUM in driving predictions.

Task 5: Conclusion and Future Work (20%)

Summary of Findings:

- Summarise model performance across all classifiers and explain which performed best and why.
- Highlight key insights from the SHAP or feature importance analysis, particularly around risk-related and payment behaviour features.

Limitations:

- Discuss dataset constraints and address any potential demographic bias and its implications for real-world financial risk assessment.
- Reflect on the impact of missing values and the choices made to handle them.

Future Work:

- Based on the limitations identified and the results obtained, propose at least **three concrete directions** for improving the current approach. These may relate to the modelling strategy, the dataset itself, the handling of class imbalance, model fairness, or deployment considerations.
- For each suggestion, briefly explain **why** it would be beneficial and what problem it addresses.
- Suggestions should be grounded in your findings.
- Avoid **generic statements** and instead connect each proposal directly to something observed during your analysis or evaluation.

Submission Guidelines:

Prepare a report showcasing your AI diagnostic tool development and model interpretation integration:

- Include screenshots or embedded plots generated using Python libraries to illustrate data

preprocessing steps, model architecture, training progress and evaluation metrics.

- Ensure all code is well-commented and includes instructions for replication.
- Ensure that your report is clear, well-organized, and visually appealing.
- Prepare the report using the **BSBI assignment template** available in Canvas.
- Upload your submission as a single file (PDF or Doc) on the BSBI portal.
- Python scripts or Jupyter notebooks, preferably uploaded to a repository platform like GitHub with a shared link.
- Ensure that all sources are cited using the **Harvard** Referencing System.
- Submit your assignment electronically by the specified deadline.

Students are encouraged to explore additional reputable sources in the digital library beyond the list provided to strengthen their individual research and critical analysis skills. The unit lecturer can help with recommending additional literature.

The criteria for successfully passing the units with two assessment components require passing both components successfully. Passing marks for BSc and MA/MSc levels are different and are stated at the end of the assignment brief. Please find the examples below:

- **Example 1** (e.g., pass mark: 50)
 - Component 1 (50% weighting) - 55
 - Component 2 (50% weighting) - 55
 - Final average: 55
 - Result: **Pass (both components passed and therefore the unit is passed)**
- **Example 2** (e.g., pass mark: 50)
 - Component 1 (50% weighting) – 80
 - Component 2 (50% weighting) – 30
 - Final average: 55
 - Result: **Fail (because although the average is above 50, one of the components is a failure)**

GRADING DESCRIPTORS: LEVEL 7

EXPERIMENTATION & INNOVATION								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Deals with complex issues both systematically and creatively demonstrating self-direction and originality in tackling and solving problems	Little to no ability to use techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions.	Low utilisation of established techniques to deal with complex issues systematically (including those of ethics and sustainability) and creatively to solve problems and/or make decisions, but with limitations in techniques or approach.	Limited research or advanced scholarship to their area of study by using a range of information and established and advanced techniques	Competent understanding of solving problems, through own research or advanced scholarship displaying a comprehensive understanding of established and advanced techniques	Good understanding of solving problems through own research and advanced scholarship critically selecting and displaying a comprehensive understanding of established and advanced techniques.	Very Good problem-solving skills displaying a comprehensive understanding of techniques applicable to their own research or advanced scholarship	Excellent range of extremely well-developed problem-solving displaying an understanding of techniques applicable to their own research or advanced scholarship beyond which is taught.	Exceptional problem-solving skills with sophisticated evaluation and application of a wide range of advanced information and techniques to undertake projects.
Comprehensive understanding of techniques applicable to their own research or advanced scholarship	Little to no understanding of techniques applicable to their own research or advanced scholarship or their limitations and ambiguities.	Low understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Limited understanding of key techniques applicable to their own research or advanced scholarship including their limitations and ambiguities.	Competent understanding of techniques applicable to their own research or advanced scholarship including their limitations and ambiguities	Good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Very good understanding of techniques applicable to their own research or advanced scholarship and a some understanding of more specialised techniques.	Excellent understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.	Exceptional understanding of techniques applicable to their own research or advanced scholarship and mastery of some more specialised areas.

GRADING DESCRIPTORS: LEVEL 7

RESEARCH & ANALYSIS								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Systematic understanding of knowledge, and a critical awareness of current problems and/or new insights, much of which is at, or informed by, the forefront of their academic discipline, field of study or area of professional practice	Little to no knowledge of the subject with limited breadth or depth or deficiencies in major areas or currency.	Low knowledge of the subject lacking coherence, breadth, or detail with only some reference to ideas or arguments at the forefront of any part of the subject.	Limited knowledge to deal with terminology, facts and concepts some of which is informed by the forefront of defined areas of the subject.	Competent knowledge of ideas or arguments at the forefront of any part of the subject sufficient to deal with current issues in the discipline, generally more descriptive than critical or analytical.	Good knowledge of ideas or arguments at the forefront of any part of the subject showing a clear, critical insight into the discipline as whole and current issues/problems.	Very good knowledge of ideas or arguments at the forefront of the subject some of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Excellent knowledge of ideas or arguments at the forefront of the subject many of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.	Exceptional knowledge of ideas or arguments at the forefront of the subject most of which are significantly beyond what has been taught and show a critical insight into the discipline and current issues/problems.
Conceptual understanding that enables the student to display originality in the application of knowledge	Little to no conceptual understanding or argument and a focus on descriptive explanations which do not comment on arguments of others or alternative views.	Low conceptual understanding and arguments are weak or poorly constructed, and the work does not critically evaluate the arguments of others or consider alternative views.	Limited conceptual understanding and argument construction with critical evaluation of alternative views or comment on advanced scholarship.	Competent conceptual understanding and argument construction with critical evaluation of a range of views and consistent engagement with advanced scholarship.	Good conceptual understanding which critically evaluate and synthesise other views and information with a thoughtful interpretation of advanced scholarship.	Very good conceptual understanding which systematically synthesises a wide range of views with a critical insight into advanced scholarship.	Excellent conceptual understanding which critically apply a wide range of views through a perceptive use of advanced scholarship.	Exceptional conceptual understanding of publishable quality with systematic engagement and usage of advanced scholarship.

GRADING DESCRIPTORS: LEVEL 7

ENGAGING WITH PRACTICE								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Practical understanding of how established techniques of research and enquiry are used to create and interpret knowledge in the discipline	Little to no evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Low evidence of background investigation, analysis, research, enquiry, ethical awareness, and/or study.	Limited background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques, with the ability to extract relevant points.	Competent investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and can critically appraise and use academic sources.	Good background investigation, analysis, research, enquiry, ethical awareness, and/or study using established techniques accurately, and possesses a well-developed ability to critically appraise a wide range of sources.	Very good, independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Excellent independent, extensive and appropriate investigation, analysis, research, enquiry, ethical awareness, and/or study well beyond the usual range, and critically evaluates this to advance the work and/or direct arguments.	Exceptional investigation, analysis, research, enquiry, ethical awareness, and/or study which demonstrates carefully considered depth and breadth and critically synthesises this to advance the work and/or direct arguments.
Originality in the application of knowledge	Little to no technical, creative or artistic skills related to their area of study.	Low technical, creative or artistic skills related to their area of study.	Limited technical, creative or artistic skills required for area of study.	Competent technical, creative or artistic skills required for area of study.	Good technical, creative or artistic skills required for area of study.	Very good range of technical, creative or artistic skills.	Excellent range of technical, creative or artistic skills	Exceptional range of technical, creative or artistic skills
Independently advance your own knowledge and understanding, and to develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.

GRADING DESCRIPTORS: LEVEL 7

REALISATION & COMMUNICATION								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Communicate information, ideas, problems and solutions to both specialist and non-specialist audiences.	Little to no clarity in the communication of ideas, problems and solutions to audiences.	Low clarity in the communication of ideas, problems and solutions to audiences.	Limited clarity in the communication of ideas, problems and solutions to audiences.	Competent communication of ideas, problems and solutions to audiences.	Good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Very good, confident and clear communication of ideas, problems and solutions to audiences in a range of means / media.	Excellent communication of ideas, problems and solutions to audiences in a range of means / media.	Exceptional communication of ideas, problems and solutions to audiences in a range of means / media.

GRADING DESCRIPTORS: LEVEL 7

PERSONAL & PROFESSIONAL CONNECTIVITY								
	FAIL			PASS				
Threshold Criteria	0-29%	30-39%	40-49%	50-59%	60-69%	70-79%	80-89%	90-100%
Independently advance your own knowledge and understanding, and develop new skills to a high level.	Little to no contribution to group activity and/or undertaking further training at a high/advanced level.	Low contribution to group activity and/or undertaking further training at a high/advanced level.	Limited contribution to group activity and/or undertaking further training at a high/advanced level.	Competent contribution to group activity and/or independently undertakes further training at a high/advanced level.	Good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Very good contribution to group activity and/or independently undertakes further training at a high/advanced level with an understanding of team roles	Excellent contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and leadership	Exceptional contribution to group activity and/or independently undertakes further training at a high/advanced level with teamwork and strong leadership.
Qualities and transferable skills necessary for employment requiring: (a) the exercise of initiative, ethical and personal responsibility (b) decision-making in complex and unpredictable contexts	Little to no ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Low ability to manage learning and/or exercise initiative, ethical and personal responsibility and/or decision-making in complex and unpredictable situations	Limited ability to manage learning and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Competent ability to manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Very good ability to systematically manage learning, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations.	Excellent ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations	Exceptional ability to manage learning on own initiative, and exercise initiative, ethical and personal responsibility, and decision-making in complex and unpredictable situations
	Little to no use of appropriate terminology, limited vocabulary and many errors in spelling, grammar and syntax.	Low use of appropriate terminology, with many errors in spelling, vocabulary and syntax.	Limited expression, style and appropriate vocabulary with errors in spelling, grammar and syntax which affect understanding.	Competent expression, style, and appropriate vocabulary with some errors in spelling, grammar and syntax which do not affect understanding.	Good expression, style and appropriate vocabulary with some errors in spelling, grammar and syntax.	Very good expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Excellent expression, style and appropriate vocabulary with minimal errors in spelling, grammar and syntax.	Exceptional expression, style and appropriate vocabulary with no errors in spelling, grammar and syntax.

GRADING DESCRIPTORS: LEVEL 7

	Little to no evidence of basic numeracy or digital literacy, hardware and software skills	Low evidence of basic numeracy or digital literacy, hardware and software skills competency.	Limited evidence of numeracy or digital literacy, hardware and software skills competency.	Adequate evidence of numeracy or digital literacy, hardware and software skills competency.	Good evidence of numeracy or digital literacy, hardware and software skills competency.	Very good evidence of numeracy or digital literacy, hardware and software skills	Excellent evidence of numeracy or digital literacy, hardware and software skills competency.	Exceptional evidence of numeracy or digital literacy, hardware and software skills competency.
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	competency.					competency.		
	Does not demonstrate achievement of professional competence when assessed against the requirements of a professional, statutory or regulatory body (PSRB).			The student has demonstrated achievement of professional competence when assessed against the requirements of a PSRB.				
	Inaccurate use of terminology with limited vocabulary and many errors in spelling, grammar and syntax. Inaccurate terminology, with many errors in spelling, vocabulary and syntax.			The student has adhered to the appropriate rules and/or conventions set by regulators or the industry.				